Lab Assignment 17.5

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Task -1:

Prompt: Develop a Python program to preprocess a loan dataset by handling missing values, encoding categorical variables, scaling numerical features, and splitting the dataset into training and testing sets. Use pandas for data handling and scikit-learn for scaling and train-test splitting. The program should prepare the dataset for machine learning model training.

Assignment-17.5.py

Assignment-17.5.py > ...

```python
# Task- 1 :
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
# Sample dataset
data = {
    'loan_amount': [10000, 15000, None, 20000, 25000],
    'credit_score': [700, 750, 720, None, 680],
    'loan_status': ['Approved', 'Rejected', 'Approved', 'Rejected', 'Approved']
}
df = pd.DataFrame(data)
print("📊 Original Data:")
print(df)
# Handle missing values
df['loan_amount'] = df['loan_amount'].fillna(df['loan_amount'].mean())
df['credit_score'] = df['credit_score'].fillna(df['credit_score'].median())
print("\n✅ After handling missing values:")
print(df)
# Encode loan_status
df['loan_status'] = df['loan_status'].map({'Approved': 1, 'Rejected': 0})
print("\n🔢 After encoding loan_status:")
print(df)
# Separate features and target
X = df[['loan_amount', 'credit_score']]
y = df['loan_status']
# Scale numerical features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("\n📏 Scaled Features:")
print(X_scaled)
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
print("\n📦 Training Data:")
print(X_train)
print(y_train)
print("\n🧪 Testing Data:")
print(X_test)
print(y_test)
```

OUTPUT:

```
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:
📊 Original Data:
   loan_amount  credit_score loan_status
0      10000.0         700.0    Approved
1      15000.0         750.0    Rejected
2          NaN         720.0    Approved
3      20000.0           NaN    Rejected
4      25000.0         680.0    Approved

✅ After handling missing values:
   loan_amount  credit_score loan_status
0      10000.0         700.0    Approved
1      15000.0         750.0    Rejected
2      17500.0         720.0    Approved
3      20000.0         710.0    Rejected
4      25000.0         680.0    Approved

🔢 After encoding loan_status:
   loan_amount  credit_score  loan_status
0      10000.0         700.0            1
1      15000.0         750.0            0
2      17500.0         720.0            1
3      20000.0         710.0            0
4      25000.0         680.0            1

📏 Scaled Features:
[[-1.5        -0.51832106]
 [-0.5         1.64135001]
 [ 0.          0.34554737]
 [ 0.5        -0.08638684]
 [ 1.5        -1.38218948]]

📦 Training Data:
[[ 1.5        -1.38218948]
 [ 0.          0.34554737]
 [-1.5        -0.51832106]
 [ 0.5        -0.08638684]]
4    1
2    1
0    1
3    0
Name: loan_status, dtype: int64

✏️ Testing Data:
[[-0.5         1.64135001]]
1    0
Name: loan_status, dtype: int64
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding>
```

Justification: This task demonstrates data preprocessing techniques required before training a machine learning model. Missing values are handled using mean and median methods, categorical data is encoded into numerical format, and numerical features are scaled using StandardScaler. Finally, the dataset is split into training and testing sets to prepare it for machine learning algorithms. This improves model accuracy and data quality.

Task 2:

Prompt: Create a Python program to clean a social media user dataset by removing duplicate records, handling missing values in numerical columns, and preparing the dataset for further analysis. Use pandas for data manipulation and ensure the dataset is cleaned and structured properly.

```python
# Task- 2:
import pandas as pd
# Sample dataset
data = {
    'user_id': [1, 2, 3, 4, 5, 1],  # Duplicate user_id
    'username': ['user1', 'user2', 'user3', 'user4', 'user5', 'user1'],
    'age': [25, 30, None, 22, 28, 25],
    'gender': ['Male', 'Female', 'Female', None, 'Male', 'Male'],
    'platform': ['Instagram', 'Facebook', 'Twitter', 'Instagram', None, 'Instagram'],
    'posts_count': [100, 150, 200, None, 250, 100],
    'likes': [1000, 1500, None, 2000,   2500, 1000],
    'engagement_level': ['High', 'Medium', 'Low', 'Medium', None   ,
                         'High']
}
df = pd.DataFrame(data)
print("📊 Original Data:")
print(df)
# Remove duplicate user profiles based on user_id
df = df.drop_duplicates(subset='user_id')
print("\n✅ After removing duplicates:")
print(df)
# Handle missing values in numerical columns
df['age'] = df['age'].fillna(df['age'].mean())
df['posts_count'] = df['posts_count'].fillna(df['posts_count'].mean())
df['likes'] = df['likes'].fillna(df['likes'].mean())
print("\n✅ After handling missing values in numerical columns:")
print(df)
```

Output:

```
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:/Users/chara/AppData/Local/Programs/
📊 Original Data:
   user_id username   age  gender   platform  posts_count   likes engagement_level
0        1    user1  25.0    Male  Instagram        100.0  1000.0             High
1        2    user2  30.0  Female   Facebook        150.0  1500.0           Medium
2        3    user3   NaN  Female    Twitter        200.0     NaN              Low
3        4    user4  22.0     NaN  Instagram          NaN  2000.0           Medium
4        5    user5  28.0    Male        NaN        250.0  2500.0              NaN
5        1    user1  25.0    Male  Instagram        100.0  1000.0             High

✅ After removing duplicates:
   user_id username   age  gender   platform  posts_count   likes engagement_level
0        1    user1  25.0    Male  Instagram        100.0  1000.0             High
1        2    user2  30.0  Female   Facebook        150.0  1500.0           Medium
2        3    user3   NaN  Female    Twitter        200.0     NaN              Low
3        4    user4  22.0     NaN  Instagram          NaN  2000.0           Medium
4        5    user5  28.0    Male        NaN        250.0  2500.0              NaN

✅ After handling missing values in numerical columns:
   user_id username    age  gender   platform  posts_count   likes engagement_level
0        1    user1  25.00    Male  Instagram        100.0  1000.0             High
1        2    user2  30.00  Female   Facebook        150.0  1500.0           Medium
2        3    user3  26.25  Female    Twitter        200.0  1750.0              Low
3        4    user4  22.00     NaN  Instagram        175.0  2000.0           Medium
4        5    user5  28.00    Male        NaN        250.0  2500.0              NaN
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding>
```

Justification: This task focuses on data cleaning and preprocessing techniques such as removing duplicate records and handling missing values. Cleaning data improves data quality and ensures accurate analysis and machine learning results. Using pandas makes it easy to manipulate and preprocess large datasets efficiently. This step is essential in real-world data analysis and data science workflows.

Task 3:

Prompt: Develop a Python program to preprocess weather data by converting date columns to datetime format, handling missing values, performing feature engineering, and normalizing numerical values. Use pandas for data processing and MinMaxScaler for normalization.

```python
# Task-3:
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
# Sample dataset
data = {
    'date': ['2024-01-01', '2024-02-01', '2024-03-01', '2024-04-01', '2024-05-01'],
    'temperature': [30, 28, None, 25, 22],
    'humidity': [80, None, 75, 70, 65],
    'rainfall': [5, 10, 15, None, 20]
}
df = pd.DataFrame(data)
print("📊 Original Data:")
print(df)
# Convert date column to datetime format
df['date'] = pd.to_datetime(df['date'])
print("\n✅ After converting date to datetime format:")
print(df)
# Handle missing values in temperature and humidity
df['temperature'] = df['temperature'].fillna(df['temperature'].mean())
df['humidity'] = df['humidity'].fillna(df['humidity'].mean())
print("\n✅ After handling missing values in temperature and humidity:")
print(df)
# Feature engineering: Create season and week_number columns
df['season'] = df['date'].dt.month % 12 // 3 + 1  # 1: Winter, 2: Spring, 3: Summer, 4: Fall
df['week_number'] = df['date'].dt.isocalendar().week
print("\n✅ After feature engineering:")
print(df)
# Normalize rainfall values using MinMaxScaler
scaler = MinMaxScaler()
df['rainfall_normalized'] = scaler.fit_transform(df[['rainfall']])
print("\n✅ After normalizing rainfall values:")
print(df)
```

Output:

```
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:/Users/chara/AppData/Local/Programs/
📊 Original Data:
         date  temperature  humidity  rainfall
0  2024-01-01         30.0      80.0       5.0
1  2024-02-01         28.0       NaN      10.0
2  2024-03-01          NaN      75.0      15.0
3  2024-04-01         25.0      70.0       NaN
4  2024-05-01         22.0      65.0      20.0

✅ After converting date to datetime format:
        date  temperature  humidity  rainfall
0 2024-01-01         30.0      80.0       5.0
1 2024-02-01         28.0       NaN      10.0
2 2024-03-01          NaN      75.0      15.0
3 2024-04-01         25.0      70.0       NaN
4 2024-05-01         22.0      65.0      20.0

✅ After handling missing values in temperature and humidity:
        date  temperature  humidity  rainfall
0 2024-01-01        30.00      80.0       5.0
1 2024-02-01        28.00      72.5      10.0
2 2024-03-01        26.25      75.0      15.0
3 2024-04-01        25.00      70.0       NaN
4 2024-05-01        22.00      65.0      20.0

✅ After feature engineering:
        date  temperature  humidity  rainfall  season  week_number
0 2024-01-01        30.00      80.0       5.0       1            1
1 2024-02-01        28.00      72.5      10.0       1            5
2 2024-03-01        26.25      75.0      15.0       2            9
3 2024-04-01        25.00      70.0       NaN       2           14
4 2024-05-01        22.00      65.0      20.0       2           18

✅ After normalizing rainfall values:
        date  temperature  humidity  rainfall  season  week_number  rainfall_normalized
0 2024-01-01        30.00      80.0       5.0       1            1             0.000000
1 2024-02-01        28.00      72.5      10.0       1            5             0.333333
2 2024-03-01        26.25      75.0      15.0       2            9             0.666667
3 2024-04-01        25.00      70.0       NaN       2           14                  NaN
4 2024-05-01        22.00      65.0      20.0       2           18             1.000000
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding>
```

Justification: This task demonstrates feature engineering and data preprocessing techniques used in data science and machine learning. Converting date columns allows extraction of useful features like season and week

number. Missing values are handled to maintain data consistency, and normalization ensures numerical values are scaled properly for machine learning models. These preprocessing steps improve model performance and data analysis accuracy.

Task -4:

Prompt: Create a Python program to preprocess a movie dataset by removing duplicate records, handling missing values, encoding categorical variables, and normalizing rating values. Use pandas for data cleaning and MinMaxScaler for normalization.

```python
# Task-4 :
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
# Sample dataset
data = {
    'movie_id': [1, 2, 3, 4, 5, 1],  # Duplicate movie_id
    'title': ['Jersey', 'Eega', 'Pokiri', 'ok jaanu', 'Dhurandhar', 'HI nanna'],
    'genre': ['Family', 'Drama', None, 'Rom-Com', 'Action', 'Family'],
    'language': ['Telugu', 'Telugu', 'Telugu', None, 'Telugu', 'Telugu'],
    'rating': [4.5, None, 3.0, 4.0, 5.0, 4.5]
}
df = pd.DataFrame(data)
print("📊 Original Data:")
print(df)
# Remove duplicate movie entries based on movie_id
df = df.drop_duplicates(subset='movie_id')
print("\n✅ After removing duplicates:")
print(df)
# Handle missing values in rating and genre
df['rating'] = df['rating'].fillna(df['rating'].mean())
df['genre'] = df['genre'].fillna(df['genre'].mode()[0])
print("\n✅ After handling missing values in rating and genre:")
print(df)

# Encode categorical variables genre and language
df['genre_encoded'] = df['genre'].astype('category').cat.codes
df['language_encoded'] = df['language'].astype('category').cat.codes
print("\n✅ After encoding genre and language:")
print(df)
# Scale rating values between 0 and 1 using MinMaxScaler
scaler = MinMaxScaler()
df['rating_normalized'] = scaler.fit_transform(df[['rating']])
print("\n✅ After normalizing rating values:")
print(df)
```

Output:

```
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:/Users/chara/AppData/Local/Programs/Python/Python313
📊 Original Data:
   movie_id       title    genre language  rating
0         1      Jersey   Family   Telugu     4.5
1         2        Eega    Drama   Telugu     NaN
2         3      Pokiri      NaN   Telugu     3.0
3         4    ok jaanu  Rom-Com      NaN     4.0
4         5  Dhurandhar   Action   Telugu     5.0
5         1    HI nanna   Family   Telugu     4.5

✅ After removing duplicates:
   movie_id       title    genre language  rating
0         1      Jersey   Family   Telugu     4.5
1         2        Eega    Drama   Telugu     NaN
2         3      Pokiri      NaN   Telugu     3.0
3         4    ok jaanu  Rom-Com      NaN     4.0
4         5  Dhurandhar   Action   Telugu     5.0

✅ After handling missing values in rating and genre:
   movie_id       title    genre language  rating
0         1      Jersey   Family   Telugu   4.500
1         2        Eega    Drama   Telugu   4.125
2         3      Pokiri   Action   Telugu   3.000
3         4    ok jaanu  Rom-Com      NaN   4.000
4         5  Dhurandhar   Action   Telugu   5.000

✅ After encoding genre and language:
   movie_id       title    genre language  rating  genre_encoded  language_encoded
0         1      Jersey   Family   Telugu   4.500              2                 0
1         2        Eega    Drama   Telugu   4.125              1                 0
2         3      Pokiri   Action   Telugu   3.000              0                 0
3         4    ok jaanu  Rom-Com      NaN   4.000              3                -1
4         5  Dhurandhar   Action   Telugu   5.000              0                 0

✅ After normalizing rating values:
   movie_id       title    genre language  rating  genre_encoded  language_encoded  rating_normalized
0         1      Jersey   Family   Telugu   4.500              2                 0             0.7500
1         2        Eega    Drama   Telugu   4.125              1                 0             0.5625
2         3      Pokiri   Action   Telugu   3.000              0                 0             0.0000
3         4    ok jaanu  Rom-Com      NaN   4.000              3                -1             0.5000
4         5  Dhurandhar   Action   Telugu   5.000              0                 0             1.0000
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding>
```

Justification: This task demonstrates data preprocessing techniques including duplicate removal, missing value handling, categorical encoding, and data normalization. Encoding categorical variables converts text data into numerical format for machine learning algorithms. Normalization scales rating values between 0 and 1 for better model performance. These preprocessing steps help prepare the dataset for machine learning and data analysis tasks.